

VESSEL ANALYSIS REPORT

vaf(%)	branch points count	tip point count	vein count	total vein length	average vein length
1.496	75.0	237.0	228.0	4230.301	18.554

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
1	[0,0]	[2,4]	3.828	TIP	BRANCH
2	[2,4]	[22,5]	19.414	BRANCH	BRANCH
3	[0,11]	[2,4]	6.828	TIP	BRANCH
4	[24,1]	[22,5]	3.828	TIP	BRANCH
5	[22,5]	[81,5]	69.941	BRANCH	BRANCH
6	[77,0]	[81,5]	6.243	TIP	BRANCH
7	[134,4]	[118,6]	15.828	TIP	BRANCH
8	[108,1]	[118,6]	10.657	TIP	BRANCH
9	[120,25]	[118,6]	20.899	TIP	BRANCH
10	[301,6]	[304,6]	2.0	TIP	BRANCH
11	[306,2]	[304,6]	3.414	TIP	BRANCH
12	[304,6]	[314,17]	14.314	BRANCH	BRANCH
13	[258,1]	[250,15]	16.314	TIP	BRANCH
14	[250,15]	[228,24]	24.728	BRANCH	BRANCH
15	[253,19]	[250,15]	3.828	TIP	BRANCH
16	[314,17]	[314,22]	4.0	BRANCH	BRANCH
17	[318,9]	[314,17]	8.243	TIP	BRANCH
18	[424,0]	[415,17]	19.728	TIP	BRANCH
19	[388,17]	[415,17]	26.828	TIP	BRANCH
20	[418,25]	[415,17]	8.243	TIP	BRANCH
21	[314,22]	[324,32]	15.485	BRANCH	BRANCH
22	[318,41]	[314,22]	43.456	TIP	BRANCH
23	[227,33]	[228,24]	8.414	TIP	BRANCH
24	[217,1]	[228,24]	27.556	TIP	BRANCH
25	[55,21]	[54,25]	3.414	TIP	BRANCH
26	[57,26]	[54,25]	2.414	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
27	[54,25]	[30,48]	37.799	BRANCH	BRANCH
28	[324,32]	[326,32]	1.0	BRANCH	BRANCH
29	[323,34]	[324,32]	1.414	TIP	BRANCH
30	[339,12]	[326,32]	24.385	TIP	BRANCH
31	[343,43]	[326,32]	20.142	TIP	BRANCH
32	[67,36]	[69,40]	3.828	TIP	BRANCH
33	[76,39]	[69,40]	6.414	TIP	BRANCH
34	[62,61]	[69,40]	24.314	TIP	BRANCH
35	[178,16]	[198,43]	34.284	TIP	BRANCH
36	[198,43]	[204,46]	6.243	BRANCH	BRANCH
37	[198,43]	[171,50]	28.899	BRANCH	BRANCH
38	[0,21]	[11,45]	27.556	TIP	BRANCH
39	[11,45]	[30,48]	20.071	BRANCH	BRANCH
40	[0,50]	[11,45]	12.071	TIP	BRANCH
41	[206,45]	[204,46]	1.414	TIP	BRANCH
42	[204,46]	[204,51]	4.0	BRANCH	BRANCH
43	[52,55]	[30,48]	23.899	TIP	BRANCH
44	[171,50]	[171,52]	1.0	BRANCH	BRANCH
45	[155,21]	[171,50]	35.385	TIP	BRANCH
46	[195,50]	[204,51]	8.414	TIP	BRANCH
47	[204,51]	[229,68]	30.627	BRANCH	BRANCH
48	[171,52]	[130,95]	66.012	BRANCH	BRANCH
49	[181,61]	[171,52]	12.899	TIP	BRANCH
50	[120,55]	[102,55]	17.828	TIP	BRANCH
51	[101,19]	[102,55]	36.657	TIP	BRANCH
52	[102,55]	[73,108]	64.77	BRANCH	BRANCH
53	[356,54]	[358,61]	6.828	TIP	BRANCH
54	[367,61]	[358,61]	8.0	TIP	BRANCH
55	[358,61]	[329,73]	33.142	BRANCH	BRANCH
56	[266,67]	[276,85]	22.314	BRANCH	BRANCH
57	[266,67]	[255,71]	11.243	BRANCH	BRANCH
58	[229,68]	[255,71]	30.485	BRANCH	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
59	[229,68]	[207,88]	31.042	BRANCH	BRANCH
60	[21,70]	[33,71]	11.414	BRANCH	BRANCH
61	[15,74]	[21,70]	6.243	TIP	BRANCH
62	[37,64]	[33,71]	7.657	TIP	BRANCH
63	[46,73]	[33,71]	12.414	TIP	BRANCH
64	[115,70]	[108,71]	6.414	TIP	BRANCH
65	[105,67]	[108,71]	4.828	TIP	BRANCH
66	[110,81]	[108,71]	9.828	TIP	BRANCH
67	[263,86]	[255,71]	20.243	TIP	BRANCH
68	[306,61]	[306,72]	10.0	TIP	BRANCH
69	[306,72]	[329,73]	22.414	BRANCH	BRANCH
70	[306,72]	[281,87]	30.385	BRANCH	BRANCH
71	[329,73]	[295,100]	47.698	BRANCH	BRANCH
72	[66,82]	[67,84]	1.414	TIP	BRANCH
73	[67,84]	[50,105]	27.799	BRANCH	BRANCH
74	[276,85]	[281,87]	4.828	BRANCH	BRANCH
75	[276,85]	[269,88]	6.828	BRANCH	BRANCH
76	[281,87]	[295,100]	21.899	BRANCH	BRANCH
77	[205,108]	[207,88]	21.485	TIP	BRANCH
78	[205,86]	[207,88]	1.414	TIP	BRANCH
79	[265,87]	[269,88]	3.414	TIP	BRANCH
80	[265,96]	[269,88]	8.657	TIP	BRANCH
81	[394,86]	[392,94]	7.828	TIP	BRANCH
82	[424,97]	[392,94]	32.243	TIP	BRANCH
83	[384,114]	[392,94]	21.899	TIP	BRANCH
84	[117,92]	[130,95]	13.243	TIP	BRANCH
85	[130,95]	[130,100]	4.0	BRANCH	BRANCH
86	[11,90]	[17,98]	9.485	TIP	BRANCH
87	[17,98]	[26,104]	11.657	BRANCH	BRANCH
88	[17,98]	[21,108]	11.071	BRANCH	BRANCH
89	[130,100]	[149,101]	19.243	BRANCH	BRANCH
90	[108,102]	[130,100]	23.071	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
91	[295,100]	[295,103]	2.0	BRANCH	BRANCH
92	[145,91]	[149,101]	10.657	TIP	BRANCH
93	[166,102]	[149,101]	18.899	TIP	BRANCH
94	[289,129]	[295,103]	28.899	TIP	BRANCH
95	[317,116]	[295,103]	25.971	TIP	BRANCH
96	[26,104]	[50,105]	23.414	BRANCH	BRANCH
97	[26,104]	[21,108]	5.243	BRANCH	BRANCH
98	[50,105]	[73,109]	25.071	BRANCH	BRANCH
99	[19,114]	[21,108]	5.414	TIP	BRANCH
100	[76,100]	[73,108]	8.243	TIP	BRANCH
101	[73,108]	[75,119]	10.828	BRANCH	BRANCH
102	[79,119]	[75,119]	3.0	TIP	BRANCH
103	[71,121]	[75,119]	3.828	TIP	BRANCH
104	[377,126]	[379,125]	1.414	TIP	BRANCH
105	[380,123]	[379,125]	1.0	TIP	BRANCH
106	[382,141]	[379,125]	17.485	TIP	BRANCH
107	[278,131]	[276,143]	11.828	TIP	BRANCH
108	[274,144]	[276,143]	1.414	TIP	BRANCH
109	[280,148]	[276,143]	5.828	TIP	BRANCH
110	[122,158]	[128,167]	10.485	TIP	BRANCH
111	[149,165]	[128,167]	23.314	TIP	BRANCH
112	[125,171]	[128,167]	3.828	TIP	BRANCH
113	[189,161]	[206,167]	19.071	TIP	BRANCH
114	[214,166]	[206,167]	7.414	TIP	BRANCH
115	[202,169]	[206,167]	3.414	TIP	BRANCH
116	[191,184]	[193,186]	2.414	BRANCH	BRANCH
117	[191,184]	[189,188]	4.414	BRANCH	BRANCH
118	[191,184]	[192,186]	1.414	BRANCH	BRANCH
119	[192,186]	[197,186]	4.0	BRANCH	BRANCH
120	[192,186]	[189,188]	3.414	BRANCH	BRANCH
121	[195,193]	[197,186]	6.828	TIP	BRANCH
122	[203,179]	[197,186]	8.657	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
123	[167,199]	[189,188]	25.728	TIP	BRANCH
124	[110,197]	[96,200]	14.243	TIP	BRANCH
125	[96,212]	[96,200]	11.828	TIP	BRANCH
126	[96,200]	[69,204]	29.728	BRANCH	BRANCH
127	[70,194]	[69,204]	10.243	TIP	BRANCH
128	[69,204]	[44,214]	28.728	BRANCH	BRANCH
129	[34,215]	[44,214]	9.414	TIP	BRANCH
130	[44,214]	[44,224]	9.0	BRANCH	BRANCH
131	[44,224]	[52,229]	9.071	BRANCH	BRANCH
132	[28,235]	[44,224]	21.899	TIP	BRANCH
133	[62,226]	[52,229]	11.657	TIP	BRANCH
134	[44,256]	[52,229]	29.314	TIP	BRANCH
135	[430,254]	[409,229]	35.042	TIP	BRANCH
136	[94,260]	[92,260]	1.0	TIP	BRANCH
137	[91,265]	[92,260]	4.414	TIP	BRANCH
138	[91,253]	[92,260]	6.828	TIP	BRANCH
139	[0,292]	[26,282]	29.142	TIP	BRANCH
140	[26,282]	[33,316]	36.728	BRANCH	BRANCH
141	[32,281]	[26,282]	5.0	TIP	BRANCH
142	[249,280]	[247,286]	5.828	TIP	BRANCH
143	[233,282]	[247,286]	14.657	TIP	BRANCH
144	[247,286]	[257,296]	14.899	BRANCH	BRANCH
145	[371,249]	[379,290]	43.314	TIP	BRANCH
146	[388,289]	[379,290]	8.414	TIP	BRANCH
147	[377,294]	[379,290]	3.414	TIP	BRANCH
148	[257,296]	[267,303]	12.485	BRANCH	BRANCH
149	[245,322]	[257,296]	29.556	TIP	BRANCH
150	[275,299]	[267,303]	8.657	TIP	BRANCH
151	[267,303]	[272,341]	39.657	BRANCH	BRANCH
152	[17,315]	[33,316]	15.414	TIP	BRANCH
153	[53,325]	[33,316]	22.314	TIP	BRANCH
154	[452,315]	[453,325]	9.414	TIP	BRANCH

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
155	[460,322]	[453,325]	7.243	TIP	BRANCH
156	[447,337]	[453,325]	14.071	TIP	BRANCH
157	[276,339]	[272,341]	3.828	TIP	BRANCH
158	[280,358]	[272,341]	19.314	TIP	BRANCH
159	[84,330]	[78,353]	24.485	TIP	BRANCH
160	[93,367]	[78,353]	30.485	TIP	BRANCH
161	[67,392]	[78,353]	45.456	TIP	BRANCH
162	[20,378]	[13,380]	6.828	TIP	BRANCH
163	[13,384]	[13,380]	3.0	TIP	BRANCH
164	[0,369]	[13,380]	16.728	TIP	BRANCH
165	[209,418]	[206,422]	4.243	TIP	BRANCH
166	[200,419]	[206,422]	6.828	TIP	BRANCH
167	[195,465]	[206,422]	48.799	TIP	BRANCH
168	[54,461]	[83,480]	37.87	TIP	BRANCH
169	[138,507]	[83,480]	73.527	TIP	BRANCH
170	[91,473]	[83,480]	9.485	TIP	BRANCH
171	[289,0]	[287,10]	10.828	TIP	TIP
172	[179,7]	[191,20]	18.556	TIP	TIP
173	[367,11]	[383,15]	19.314	TIP	TIP
174	[148,13]	[142,41]	35.213	TIP	TIP
175	[474,13]	[470,18]	19.485	TIP	TIP
176	[371,36]	[423,39]	55.728	TIP	TIP
177	[281,41]	[281,56]	15.828	TIP	TIP
178	[454,70]	[455,83]	18.485	TIP	TIP
179	[189,75]	[183,81]	8.485	TIP	TIP
180	[452,84]	[438,94]	19.899	TIP	TIP
181	[262,99]	[246,119]	28.385	TIP	TIP
182	[34,114]	[50,129]	24.556	TIP	TIP
183	[92,120]	[91,123]	3.414	TIP	TIP
184	[269,122]	[251,129]	21.728	TIP	TIP
185	[141,124]	[147,145]	24.314	TIP	TIP
186	[21,127]	[22,135]	9.243	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
187	[32,130]	[33,158]	30.071	TIP	TIP
188	[123,130]	[103,136]	22.485	TIP	TIP
189	[210,133]	[205,153]	24.314	TIP	TIP
190	[180,141]	[161,146]	21.071	TIP	TIP
191	[4,144]	[38,208]	91.74	TIP	TIP
192	[36,148]	[55,167]	38.142	TIP	TIP
193	[94,149]	[86,173]	34.971	TIP	TIP
194	[249,154]	[253,167]	14.657	TIP	TIP
195	[261,170]	[237,197]	36.941	TIP	TIP
196	[431,170]	[448,171]	18.243	TIP	TIP
197	[16,178]	[15,194]	17.243	TIP	TIP
198	[465,194]	[484,231]	46.627	TIP	TIP
199	[137,196]	[130,228]	38.556	TIP	TIP
200	[319,197]	[329,201]	11.657	TIP	TIP
201	[2,212]	[0,222]	10.828	TIP	TIP
202	[255,246]	[253,274]	30.485	TIP	TIP
203	[221,248]	[234,262]	24.657	TIP	TIP
204	[124,253]	[118,266]	16.071	TIP	TIP
205	[138,258]	[169,266]	34.314	TIP	TIP
206	[10,263]	[2,264]	8.414	TIP	TIP
207	[196,287]	[177,292]	21.899	TIP	TIP
208	[440,318]	[416,330]	28.971	TIP	TIP
209	[182,320]	[175,340]	22.899	TIP	TIP
210	[452,341]	[430,399]	67.941	TIP	TIP
211	[15,345]	[49,348]	40.213	TIP	TIP
212	[401,347]	[370,368]	46.728	TIP	TIP
213	[142,350]	[148,354]	7.657	TIP	TIP
214	[127,375]	[128,398]	25.899	TIP	TIP
215	[81,400]	[86,420]	22.071	TIP	TIP
216	[358,402]	[363,424]	24.071	TIP	TIP
217	[227,404]	[215,412]	15.314	TIP	TIP
218	[136,413]	[146,422]	16.657	TIP	TIP

id	p1.x, p1.y	p2.x, p2.y	length	p1_type	p2_type
219	[160,418]	[151,431]	16.728	TIP	TIP
220	[40,421]	[55,421]	18.314	TIP	TIP
221	[28,428]	[0,443]	35.385	TIP	TIP
222	[115,439]	[133,451]	22.971	TIP	TIP
223	[371,451]	[372,476]	28.485	TIP	TIP
224	[15,478]	[18,481]	5.414	TIP	TIP
225	[173,481]	[200,488]	50.284	TIP	TIP
226	[245,485]	[238,499]	16.899	TIP	TIP
227	[281,498]	[309,511]	35.142	TIP	TIP
228	[50,499]	[48,510]	11.828	TIP	TIP